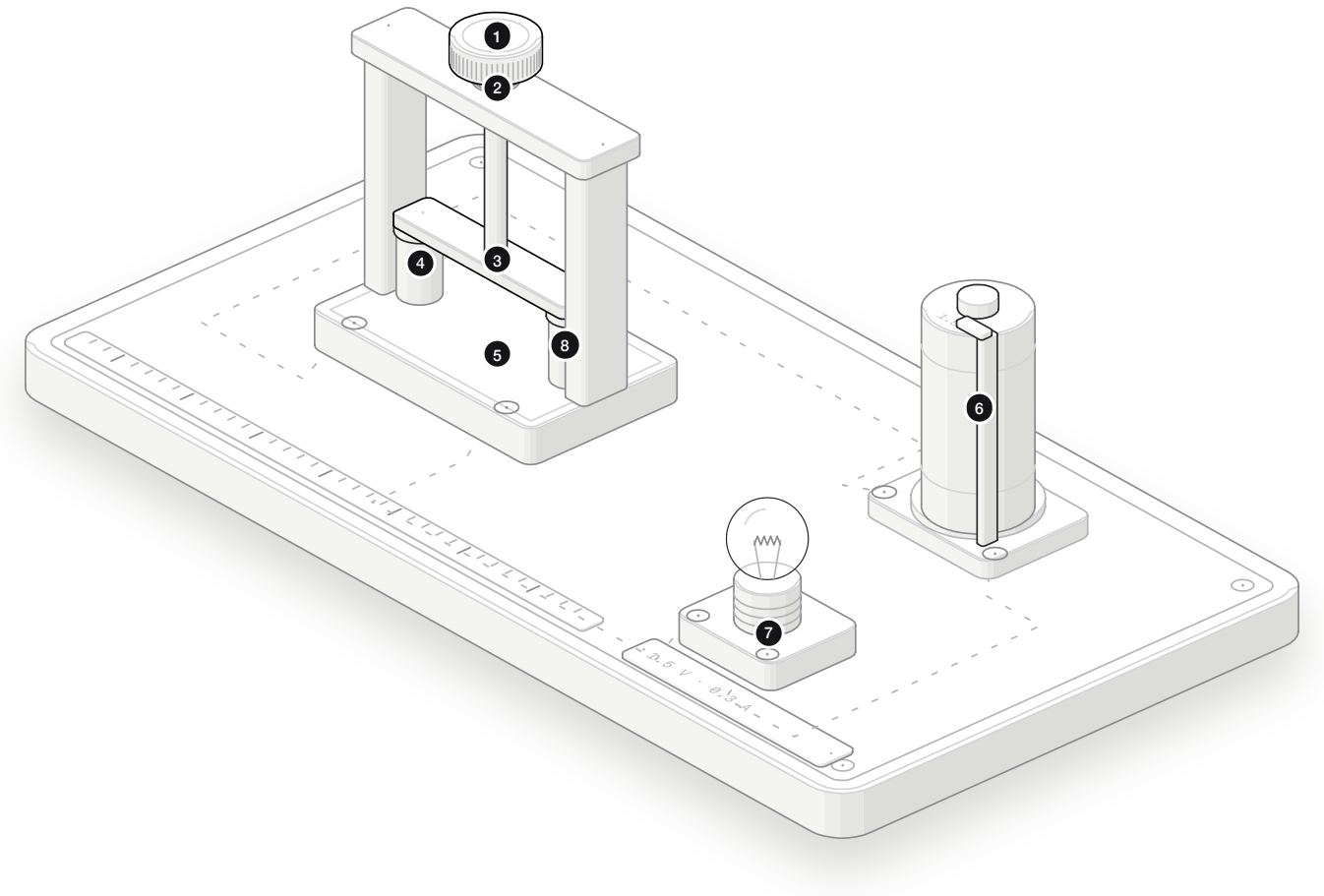
Push button

AN–003
HOW IT WORKS
A LOOP, A GAP AND A SPRING



How does a button turn a light on?

A battery can only push current round a closed loop. The button is a gap in that loop: a spring holds a copper bridge just above two contact posts. Press the cap and the bridge lands on both posts, the loop closes and the bulb lights. Let go and the spring opens the gap again.

1.5 V
FROM THE CELL

0.30 A
ROUND THE CLOSED LOOP

0.45 W
IN THE BULB

7 px
GAP THE BRIDGE CLOSES

01 **Cap**
Where your finger pushes; it carries the plunger.

02 **Return spring**
Holds the button open and lifts it when you let go.

03 **Copper bridge**
Rides on the plunger and closes the gap between the posts.

04 **Contact posts**
Two brass posts, one for each end of the gap.

05 **Frame and base**
An insulating block and a frame that guides the plunger.

06 **1.5 V cell**
Pushes current round the loop once it is closed.

07 **Bulb**
Glows when current runs through its thin filament.

08 **Wires**
Carry current from the cell, through the button, to the bulb and back.